

73-103 · CMU-Q

Principles of Macroeconomics

Fall 2026

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Meeting 2 · Measuring GDP

Measuring a Nation's Income

Wed 26 Aug

Course Roadmap

	Topic	Description	Tools	Use
Measurement	GDP	What the economy produced	Circular flow, $Y \equiv C+I+G+NX$	Read a release; Qatar energy GDP
	Prices	Persistent rise in the price level	CPI / deflator, real vs nominal	Contract, wage, peg
	Labor	Employment, unemployment, participation	Labor-force identities; short AI/labor	GCC hiring / participation
Short-run demand	Goods / multiplier	Demand determines Y	ZZ, multiplier, Blanchard 3	Fiscal impulse
	Financial markets	How i is set	Money market, Blanchard 4	QCB / Fed; $i = \bar{i}$
	IS-LM	Goods + financial together	IS, LM, policy mix, dashboards	Shock → which curve, which way
Medium run	Phillips	Inflation and unemployment together	Phillips curve, Blanchard 7–8	Wage / inflation tradeoff
Long run	Growth	Why levels differ	Production function, growth accounting	Handout 4 + R
Open	Open economy	Trade, rates, and the peg	NX, Blanchard 17–19	Shock → NX / peg (Qatar)

This meeting

1

Discussion on readings

Yen-carry article. Huang vs Amodeli.
CORE 3.1–3.4: what stuck?

2

Macroeconomic Policy and the cycle

Objectives, tools, and what
macroeconomics studies.

3

Why do we measure economic activity?

Firms, households, and the fiscal
authority all need a measure of
activity.

4

Circular flow of the economy

Income $\equiv$ expenditure. Two sectors,
then four.

5

Three methods for measuring GDP

Output (value added), income,
expenditure. Same Y.

6

Components of the GDP

C, I, G, NX.

BLOCK 1

Discussion on readings

**What did you take from the yen-carry article,
Huang vs Amodei, and CORE 3.1–3.4?**

After Monday

1 Yen-carry trade

What did the Foreign Policy article add to Monday's discussion on Yen carry trade?

2 Huang vs Amodei

After the video and the Axios piece, is your view of AI and jobs more nuanced, or more extreme?

3 CORE 3.1-3.4

What was the most interesting idea in the reading, and why?

BLOCK 2

Macroeconomics : Objectives and Scope

Objectives of Macroeconomic Policy and What does Macroeconomics Study

What are the objectives of macroeconomic policy?

Growth

Steady expansion without boom–bust cycles.

Full employment

Jobs for those who want to work, near the natural rate.

Redistribution

How the gains are shared.
Not the same object as GDP.

Price stability

Low, predictable inflation.
Preserves purchasing power.

Financial stability

A banking system that does not seize up.

Exchange-rate stability

Predictable relative prices for trade and capital.

Fiscal Policy

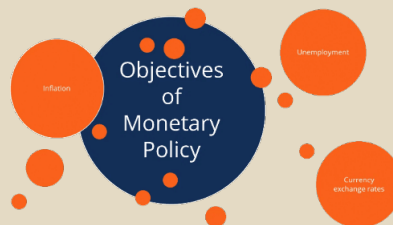
G and T. The government.

Monetary Policy

The policy rate, reserves, open-market operations.

Exchange rate Policy

Fix, float, or manage.



What does macroeconomics study?

1

Fluctuations

Expansions and contractions, the business cycle.

2

National output

The production of goods and services. GDP is the object.

3

Labor markets

Employment, unemployment, participation.

4

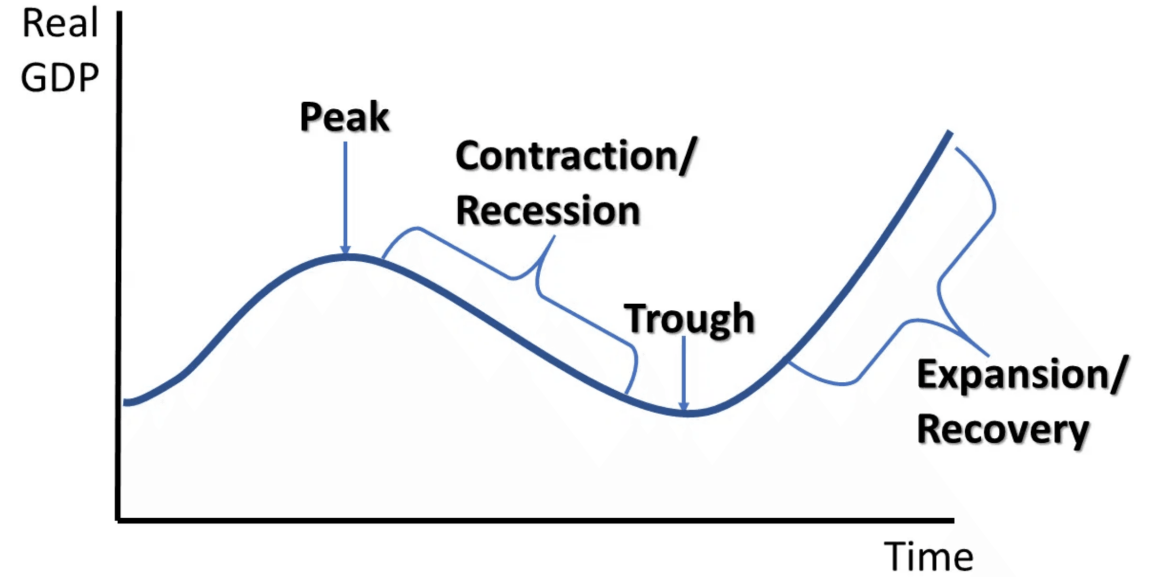
Price level

Inflation and deflation. Not one relative price.

5

Growth

The long-run path, distinct from the cycle.



The business cycle — peak, contraction, trough, expansion.

Where is the Qatari economy in this cycle today?

BLOCK 3

Why measuring economic conditions

Why does a firm, a household, and a finance ministry all need a measure of activity?

Why measuring activity matters

Prevailing conditions guide production, consumption, and public finance.

Firms

Produce

Production, inventories, hiring, and capex are set on a view of demand.

Households

Work · borrow

Whether to enter work, invest in schooling, or take on a mortgage.

Governments

Debt / GDP

taking more debt to expand social programs.

Reporting changes in activity is fundamental for decisions about production, consumption, and investment.

How changes in activity are reported

GDP

Production

The market value of output.
Today's object: the total
income of everyone in the
economy.

π

Inflation

The rate of change of the
price level.

u

Unemployment

The share of the labor force
without work.

NX

External balance

Net exports, and the
current account.

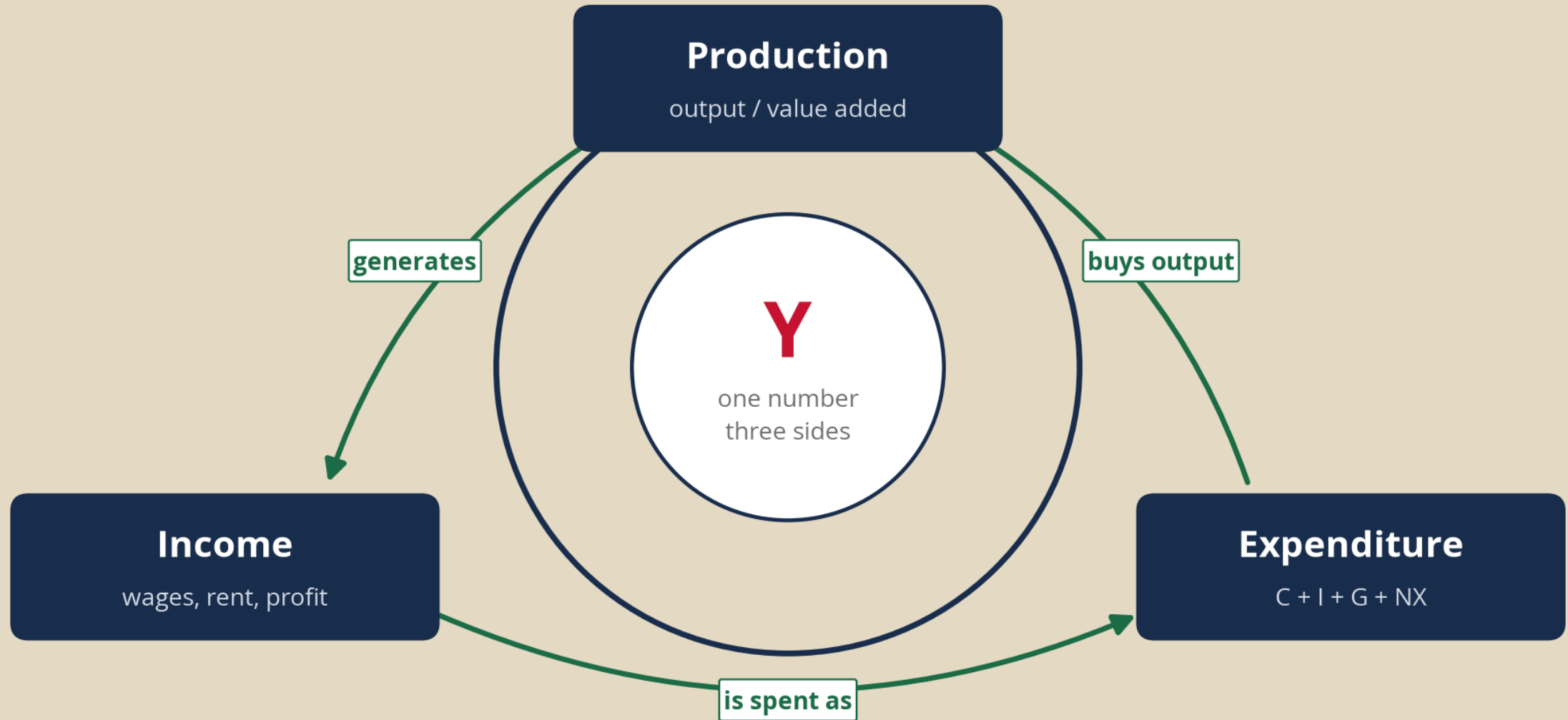
BLOCK 4

GDP Measurement and the Circular-Flow Diagram

Three (3) different ways to measure GDP

The circular flow model

The circular flow · production, income, and expenditure are the same Y



Three ways to measure GDP

If we counted everything, these are equal. Recessions: output falls, so expenditure and income fall.

Valued-added

$$Y = \sum VA_i$$

$$VA_i = P_i Q_i - \text{intermediates}_i$$

- Stops double-counting.
- The shirt, not cotton + cloth + shirt.

Income

$$Y = W + R + \text{Profits}$$

- Wages, rent, interest, and profit.
- What the owners of labor, land, and capital receive.

Expenditure

$$Y \equiv C + I + G + (X - M)$$

- True by construction.
- Inventories sit in I so output still equals spending when firms produce more than they sell.

Example: Producing a shirt, three industries

Value added at each stage = price of output – intermediate inputs



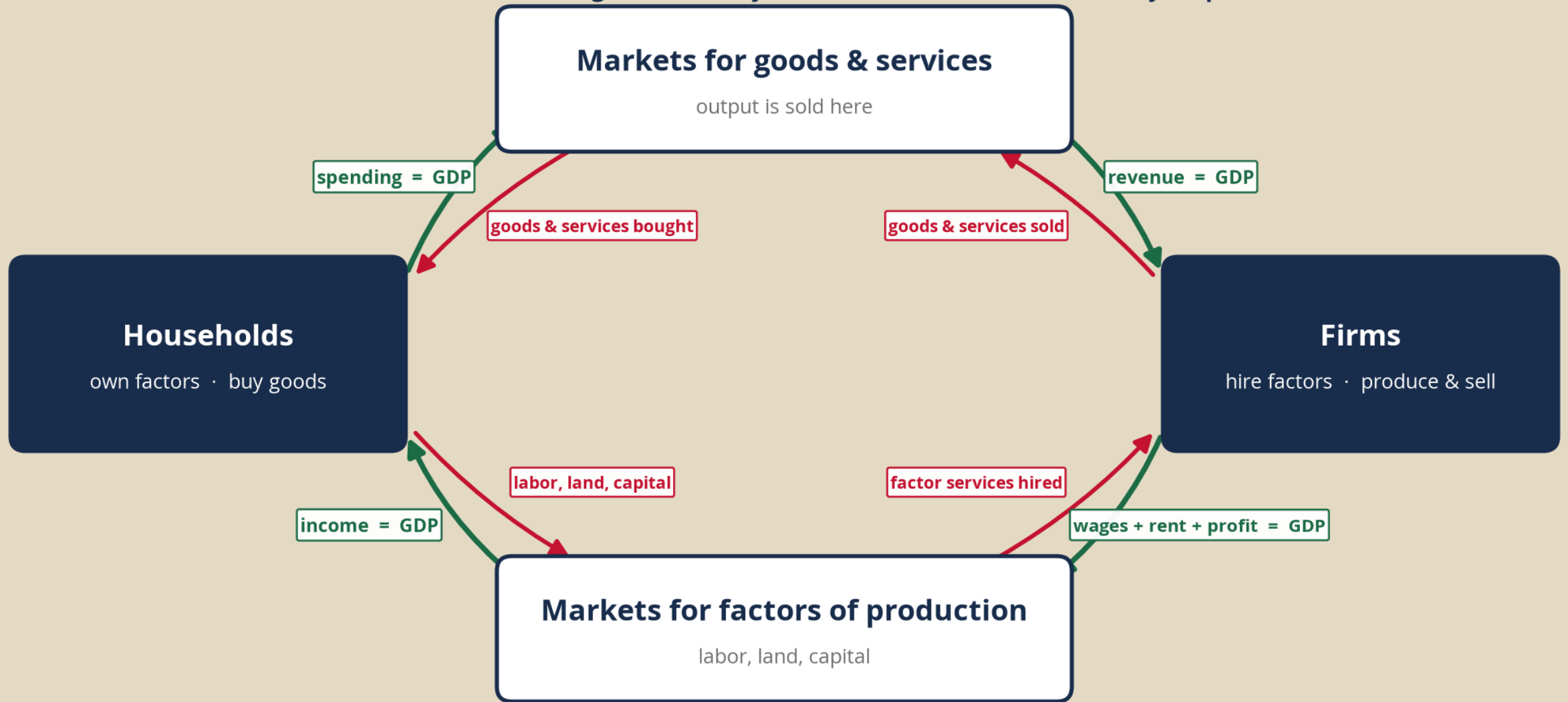
$$\text{GDP} = 50 + 30 + 20 = 100 = \text{final sale}$$

Wrong: $50 + 80 + 100 = 230$ (double-counting intermediates)

Same algebra for LNG: wellhead → liquefaction train → tanker. Do not add the three invoices.

Circular flow model: Two sectors

Two-sector circular flow • green = money • red = real • both money loops are GDP



What two sectors omit

Government

T

taxes

G

purchases

TR

transfers $\neq$
G

Pensions and unemployment insurance are TR, not G.

The retiree's spending of the check is already in C.

Financial system

S



I

Saving finances investment.

Buying a QNB share is a financial transaction — not I in the accounts.

Rest of the world

X

exports

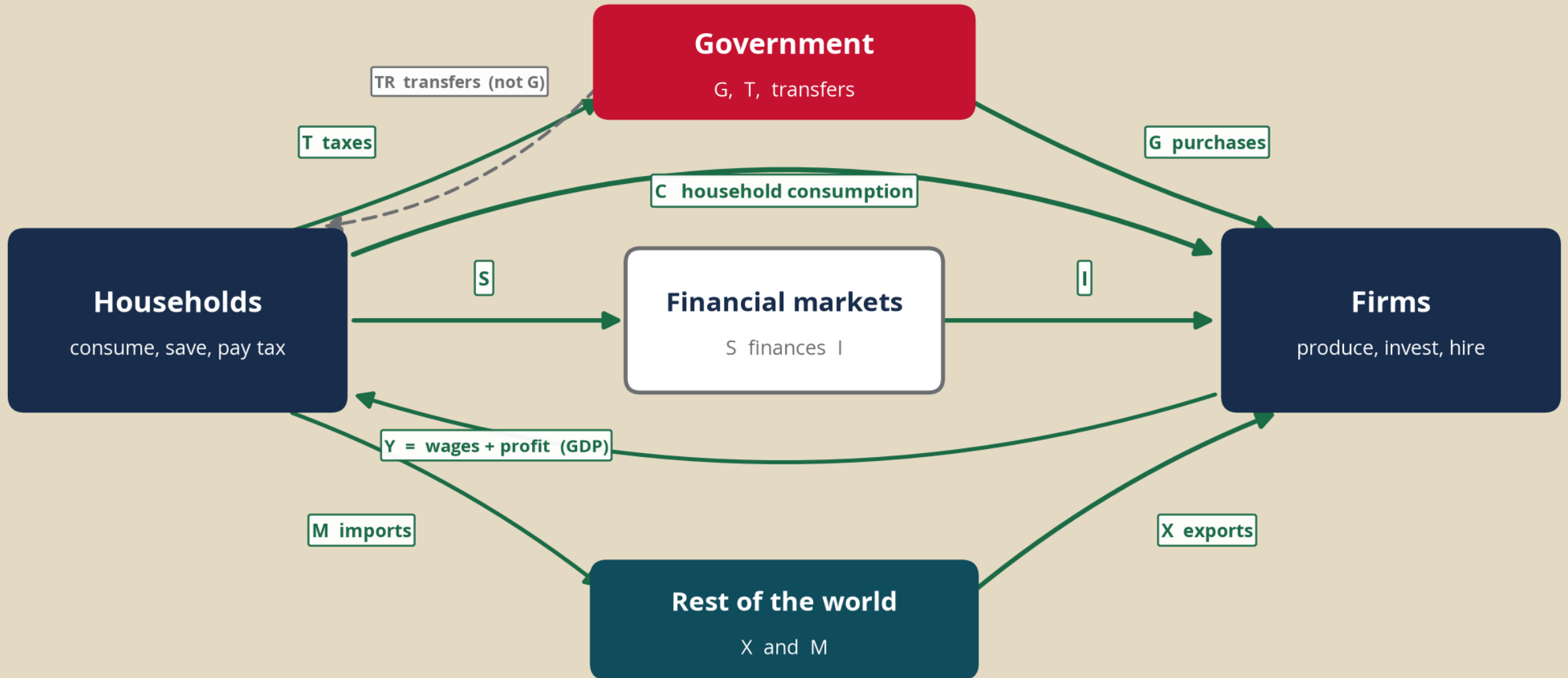
M

imports

Net factor income is how GDP and GNI come apart. Trace X in, M out.

Four-sector circular flow

Four-sector circular flow — money. Real goods and factors run the opposite way. Dashed = transfers (not G).



BLOCK 5

The components of GDP

GDP: Components of the expenditure approach

The Components of GDP

C

Consumption

Spending by households on goods & services

I

Investment

- Purchase of capital goods, structures by household and firms, and
- inventories
- Also called gross fixed capital formation

G

Government

Government purchases of goods & services

NX

Net Exports

Exports minus Imports (X – M)

$$Y = C + I + G + NX$$

Consumption (C)

Household spending on goods and services. New housing is I, not C.

Durables

Cars, appliances. Last more than a year.

Nondurables

Food, fuel, clothing.

Services

Haircuts, health, education.

Housing

Renter

Rent paid is C.

Owner

Imputed rent is C. Not the purchase price, not the mortgage.

New house

The structure is I, not C.

Investment (I)

Spending on goods that will be used to produce more goods. Also called gross fixed capital formation.

Business capital

Structures, equipment, intellectual property.

Residential

A landlord's building; a homeowner's residence.

Inventories

Goods produced but not yet sold.

Not I

Buying QNB shares or a Treasury bill is a financial transaction. It is not investment in the national accounts.

Government purchases (G)

G = purchases

Goods and services

Teachers, hospitals, the metro, a new school.

Federal, state, and local.

Government consumption expenditure and gross investment.

Transfers $\neq$ G

Not a purchase

Social Security, unemployment insurance.

Double-count if you add them

The retiree's spending of the check is already in C.

G is taken as given this period.

Net exports (NX)

X

Foreign spending on our goods and services.

—

M

The parts of C, I, and G spent on foreign output.

=

NX

X - M

Adding the components

$$Y = C + I + G + NX$$

Imports already sit inside C, I, and G. Subtracting M stops us counting foreign output as ours.

Two Very Different Economies

United States

Consumption (C)	68%
Investment (I)	18%
Government (G)	17%
Net Exports (NX)	-3%

A consumption-driven economy with a trade deficit.

Qatar

Consumption (C)	22%
Investment (I)	38%
Government (G)	12%
Net Exports (NX)	+28%

An export-driven economy (LNG/Oil) with a large trade surplus.